

MAIE5421/ARIN5201 (2026 Fall) Course Project Guidelines

HKUST

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Overview

Form groups of 5–6 students and select **one** project from the provided list of computer vision and machine learning topics. You may also propose your own topic (see the last entry).

Deliverables

Submit the following items by the corresponding deadline:

1. Project Proposal

Prepare and submit a document no more than 2 pages containing the following: your project name, group member information, project background, and a preliminary methodology with a project work timeline.

2. Technical Report

6-page CVPR-style paper (references excluded from page count) containing:

- Team member names and responsibilities
- Project timeline with milestones
- Methodology and technical approach
- Experimental results and analysis
- Conclusion and references

3. Presentation

Give a final presentation of about 10 minutes. Slides do not need to be submitted.

Requirements

- The contributions of all team members should be highly relevant to the outcomes of the project.
- Report is supposed to follow CVPR formatting guidelines. You can find the official template at <https://github.com/cvpr-org/author-kit/releases>
- Projects should address all original requirements. Implementations that handle challenges in corresponding areas successfully are welcomed and can contribute to higher scores.

1 Creative Poster Generator

1.1 Problem Background

Current AI systems struggle to generate high-quality posters from abstract concepts. Single-model approaches often fail to maintain coherence between semantic meaning and aesthetic quality. This project explores how AI agents can collaboratively transform textual ideas into compelling posters.

1.2 Project Description

Design an agent system that converts keywords into poster images. Your input is 2-5 keywords; output is a high-resolution poster. Between these points, design your own workflow with 3-4 specialized models.

1.2.1 Requirements

- Input: 2-5 keywords
- Output: 720×1280 poster image
- 3-4 specialized agents with clear roles
- Robust inter-agent communication

1.3 Expected Outcomes

A complete system that demonstrates:

- **Concept Understanding:** Interprets abstract keywords meaningfully
- **Visual Generation:** Creates aesthetically pleasing imagery
- **Layout Design:** Applies composition principles effectively
- **Style Consistency:** Maintains coherent visual language

1.4 Technical Guidance

1.4.1 Agent Design Steps

1. Decompose poster creation into logical subtasks
2. Assign specialized roles to each agent
3. Design clear communication protocols
4. Implement quality validation checks based on several evaluation criteria (e.g., output quality, instruction following, robustness, efficiency)

1.4.2 Potential Agent Roles

- Concept Expander: Keywords $\rightarrow$ creative brief
- Visual Designer: Brief $\rightarrow$ visual elements
- Layout Planner: Elements $\rightarrow$ composition
- Quality Refiner: Iterative improvement

Note: Students have complete freedom in designing the intermediate workflow between the fixed input/output endpoints. Such technical guidance only serves as a reference for your project development.

2 Multi-Subject Image Customization

2.1 Project Background

Current image customization techniques excel at learning single subjects but struggle with composing multiple custom subjects into coherent scenes. When generating images containing multiple personalized elements, models often suffer from identity bleeding, where attributes from one subject contaminate another, and poor spatial reasoning. This project addresses the challenge of faithful multi-subject generation, requiring models to maintain distinct subject identities while creating plausible interactions and compositions.

2.2 Project Description

Design a system that can generate novel images containing multiple customized subjects from separate reference images. Your solution should preserve each subject's distinct visual identity while creating a coherent composition that reflects their relationships.

2.2.1 Requirements

- **Input:** 2-4 reference images, each containing a distinct subject, plus a text prompt describing their desired interaction
- **Output:** 1024×1024 image containing all customized subjects in a coherent scene
- **Core Challenge:** Maintain clear separation of subject identities while ensuring plausible spatial relationships. You may consider diverse subjects, including objects, human portraits.
- **Evaluation:** Comprehensive analysis of identity preservation and compositional quality

2.3 Expected Outcomes

A complete pipeline that demonstrates robust performance in:

- **Identity Preservation:** Each subject keeps its distinctive features without attribute mixing
- **Spatial Coherence:** Logical placement and scaling of subjects within the scene
- **Composition Quality:** Aesthetically pleasing arrangement with appropriate lighting consistency
- **Prompt Adherence:** Faithful realization of the described interaction between subjects

2.4 Technical Guidance

2.4.1 Potential Approaches

- **Multi-Concept Fine-tuning:** Adapt single-subject methods (DreamBooth, LoRA) to handle multiple concepts simultaneously
- **Attention Control:** Implement cross-attention guidance to separate subject representations during generation
- **Cascaded Refinement:** Generate subjects individually then compose them with blending techniques

Note: Student should focus on maintaining clear separation between subject representations while ensuring the final composition appears natural and coherent. Consider both the technical implementation and the perceptual quality of your results.

3 3D Assets Generator

3.1 Project Background

The field of computer vision is rapidly advancing from 2D image generation to 3D asset creation, which is crucial for applications in augmented reality, gaming, and the metaverse. Traditionally, creating high-quality 3D models is time-consuming and skill-intensive. Recent breakthroughs in generative AI, particularly diffusion models and neural radiance fields (NeRFs), are democratizing this process [5]. This project explores how AI can transform simple inputs into detailed 3D objects, tackling exciting challenges in modern computer vision.

3.2 Project Description

Students will design a system that uses generative AI models to create 3D assets from text or image inputs. The system should produce usable 3D model files that faithfully represent input descriptions or reference images.

3.2.1 Requirements

- Input: A text prompt OR a single 2D image
- Output: A 3D model file (.obj, .glb, .fbx) with textures
- Core Methodology: Implement at least one modern 3D generation technique (e.g., Diffusion Models, NeRFs, 3D Gaussians)
- Evaluation: Final report with quantitative and qualitative analysis

3.3 Expected Outcomes

A successful project will deliver a complete pipeline and report demonstrating:

- **Geometric Quality:** Accuracy and structure of 3D shape using metrics like Hausdorff Distance
- **Visual Fidelity:** Realism and quality of applied textures and materials
- **Semantic Consistency:** Alignment between generated 3D model and input content

3.4 Technical Guidance

3.4.1 Implementation Steps

1. Select a base model and 3D representation approach
2. Preprocess input data and set up training pipeline
3. Implement chosen generation technique with optimization
4. Generate 3D assets and evaluate using specified metrics

3.4.2 Potential Approaches

- **State-of-the-art 3D Asset Models:** Hunyuan3D 2.0 by Tencent and Trellis3D by Microsoft.

3.4.3 Implementation Tips

- Use clear, well-lit images with simple backgrounds for image inputs
- Employ specific prompts describing materials, style, and proportions for text inputs
- Plan for post-processing in software like Blender for optimal quality

Note: Students have creative freedom in designing your implementation workflow between the fixed input/output requirements. This technical guidance serves as reference for project development.

4 Video Style Transfer

4.1 Project Background

Video style transfer represents an advanced extension of image style transfer techniques, aiming to apply the artistic style from a reference image to all frames of a video while preserving the original content. Unlike image processing, video style transfer faces the critical challenge of **temporal consistency** - ensuring stable, flicker-free results across frames despite object and camera motion. This technology has applications in film production, social media filters, and creative video editing.

4.2 Project Description

Students will develop a video style transfer system that transforms input videos into stylized versions matching a reference image's artistic characteristics. The core challenge involves maintaining both **spatial style quality** and **temporal coherence** across frames.

4.2.1 Requirements

- **Input:** A short video (3-5 seconds) and one reference style image
- **Output:** A stylized video maintaining original content with consistent artistic style
- **Core Challenge:** Implement temporal consistency mechanisms to reduce flickering and artifacts
- **Evaluation:** Comprehensive analysis of style quality, content preservation, and temporal stability

4.3 Expected Outcomes

A successful project will deliver a complete pipeline and evaluation report demonstrating performance in these key areas:

- **Style Fidelity:** Visual similarity to reference style;
- **Content Preservation:** Structural similarity index; Content loss evaluation
- **Temporal Consistency:** Flow-based consistency checks; Visual flicker assessment

4.4 Technical Guidance

4.4.1 Implementation Framework

- Utilize **PyTorch** or **TensorFlow** with pre-trained VGG networks for feature extraction
- Implement both **content loss** and **style loss** functions based on deep feature representations
- Add **temporal coherence loss** using optical flow or frame warping techniques
- Consider building upon existing video style transfer codebases

4.4.2 Practical Considerations

- Start with image-style transfer before extending to video
- Use shorter video clips initially to manage computational resources
- Experiment with different weightings between content, style, and temporal loss components
- Consider efficient optical flow algorithms (like FlowNet) for temporal regularization

Note: Students have creative freedom in designing your implementation workflow between the fixed input/output requirements. This technical guidance serves as reference for project development.

5 Intelligent Video Clip Generator

5.1 Project Background

The exponential growth of video content demands efficient tools for parsing and extracting key moments. Current AI video models excel at generating content from text but struggle with precise, prompt-based retrieval from existing long videos. This project bridges that gap by combining video understanding and conditional generation. Students will build a system that understands both the semantic content of a long video and the user's textual intent to generate concise, relevant clips.

5.2 Project Description

Students will design and implement a system that takes a long video and a text prompt as input, and outputs a shorter, coherent video clip that is highly relevant to the prompt.

5.2.1 Requirements

- **Input:** A long video (around 5 minutes) and a text prompt describing desired clip content
- **Output:** A short video clip (5-15 seconds) highly relevant to the prompt
- **Core Functionality:** Must perform temporal localization and clip generation/refinement
- **Evaluation:** Final report with quantitative and qualitative analysis

5.3 Expected Outcomes

A successful project will deliver a functional pipeline demonstrating performance in:

- **Prompt Relevance:** CLIP Score measuring text-video alignment
- **Temporal Accuracy:** Precision of segment selection within long video
- **Overall Usability:** Qualitative assessment of end-to-end performance

5.4 Technical Guidance

5.4.1 Recommended Approaches

- **Two-Stage Pipeline:**
 1. Moments Detection using video-language models
 2. Clip Refinement with conditional video generation models
- **Exploratory Approach:** End-to-end generation using text-to-video models

5.4.2 Implementation Tools

- **Video Understanding:** Pre-trained models for temporal action localization
- **Video Generation:** Open-source video generation models

5.4.3 Critical Considerations

- Maintain temporal consistency in generated clips
- Handle harder cases where multiple prompt contents can be extracted from original video.
- Balance computational efficiency with output quality

Note: Students should focus on creating a robust pipeline that effectively bridges video understanding and generation. The technical guidance serves as reference for project development.

6 AI-Generated Image Detection

6.1 Project Background

The proliferation of AI-generated images poses significant challenges for digital media authenticity and trust. Current generative models can produce highly realistic synthetic images that are increasingly difficult to distinguish from real photographs. This project addresses the critical need for reliable detection methods by leveraging deep learning to identify AI-generated visual content. Students will develop a system that analyzes image artifacts and patterns indicative of synthetic generation.

6.2 Project Description

Students will design and implement a classification system that takes an image as input and determines whether it was AI-generated or authentic, providing a confidence score for the prediction.

6.2.1 Requirements

- **Input:** Images in common formats (JPEG, PNG) with varying sources
- **Output:** Binary classification (Real/AI-Generated) with confidence probability
- **Core Challenge:** Robust classification across various generation models
- **Evaluation:** Comprehensive testing on diverse datasets with performance metrics analysis

6.3 Expected Outcomes

A successful project will deliver a reliable detection system demonstrating performance in:

- **Detection Accuracy:** Achieve and quantitatively demonstrate high performance (precision, recall, etc.) on a held-out test set comprising images
- **Generalization and Robustness:** Maintain competitive performance on datasets from **unseen or emerging** generative models and various real-world perturbations.

6.4 Technical Guidance

6.4.1 Implementation Steps

1. Construct datasets using various generative models and perform train-val-test splits.
2. Select a base model (e.g., pre-trained CNNs or Vision Transformers).
3. Design the training framework, including data preprocessing, model architecture adjustments, and objective function design.
4. Train the model and test on datasets: hold-out test set, unseen data, and perturbed data.

6.4.2 Recommended Datasets and Models

- **Primary Dataset:** ImageNet-1K
- **Generative Models:** A selection including, but not limited to, BigGAN, StyleGAN2, ProGAN, GigaGAN, LDM, Stable Diffusion series, and DALL · E.

6.4.3 Critical Considerations

- Handle diverse generative methods (GANs, Diffusion Models).
- Address dataset bias and ensure fair evaluation across image types.
- Maintain robustness against image transformations.

Note: Students should focus on creating a robust detection system that generalizes well across different generation techniques. The technical guidance serves as reference for project development.

7 Watermarking

7.1 Project Background

The increasing need for copyright protection and content authentication in digital media has driven significant interest in robust image watermarking techniques. Unlike fragile watermarks that detect tampering, robust watermarks must survive various image processing operations while remaining imperceptible. This project addresses the dual challenge of maintaining visual quality and robustness by developing an advanced watermarking system that balances these critical requirements.

7.2 Project Description

Students will design and implement a robust invisible watermarking system that embeds information into images while preserving visual quality and ensuring detectability after image transformations.

7.2.1 Requirements

- **Input:** Original images and watermark payload (binary sequence or identifier)
- **Output:** Watermarked images and extracted watermark after processing
- **Core Functionality:** Watermark embedding and extraction with robustness to image transformations
- **Evaluation:** Comprehensive analysis of imperceptibility and robustness under various image transformations

7.3 Expected Outcomes

A successful project will deliver a functional watermarking system demonstrating performance in:

- **Imperceptibility:** High PSNR/SSIM values maintaining visual quality
- **Robustness:** High watermark recovery rates after compression, cropping, and other transformations

7.4 Technical Guidance

7.4.1 Implementation Steps

1. Define the watermark format: text, bit string, image, or other custom forms.
2. Build the framework:
 - **Generation:** Embed watermarks during the image generation process.
 - **Detection:** Identify the presence of watermarks in images.
3. Train the generator and detector. Evaluate the watermark detection accuracy on generated images and assess robustness against various image transformations.

7.4.2 Recommended Models

- Stable Diffusion series

7.4.3 Critical Considerations

- Balance watermark capacity with visual quality.
- Address various attack scenarios: JPEG compression, cropping, rotation, and noise addition.

Note: Students should focus on creating a robust watermarking system that effectively balances imperceptibility and robustness. The format of the watermark can be customized, such as an image, a string, etc. The technical guidance serves as reference for project development.

8 Class-agnostic Object Counting

8.1 Project Background

Object counting is a fundamental computer vision task with wide applications in surveillance, traffic management, etc. Traditional object detection methods struggle with dense scenes and varying object scales, making accurate counting challenging. Recent density-based estimation approaches have shown promising results by predicting density maps instead of detecting individual instances. This project explores advanced counting methods that can handle complex real-world scenarios.

8.2 Project Description

Students will develop an object counting system that takes an image as input and estimates the number of target objects, with particular focus on handling dense distributions and scale variations.

8.2.1 Requirements

- **Input:** Single RGB image containing multiple instances of target objects
- **Output:** Estimated object count and corresponding density map visualization
- **Core Functionality:** Must handle varying objects, densities and scales
- **Evaluation:** Quantitative metrics and qualitative analysis on test images

8.3 Expected Outcomes

A successful project will deliver a robust counting system demonstrating performance in:

- **Counting Accuracy:** Low mean absolute error (MAE) and root mean square error (RMSE)
- **Density Map Quality:** Meaningful density distributions correlating with object locations
- **Scale Robustness:** Consistent performance across different objects, sizes and densities

8.4 Technical Guidance

8.4.1 Recommended Approaches

- **Density Map Estimation:** Utilize CNN-based models to generate spatial density maps for accurate object counting.
- **Prompt-based Counting:** Employ visual prompts (e.g., exemplar bounding boxes) to enable counting of arbitrary object categories.

8.4.2 Dataset

FSC-147 (Ranjan et al. Learning To Count Everything.)

8.4.3 Critical Considerations

- Maintain balance between counting accuracy and model generalization to unseen object categories
- Address challenges including scale variation, occlusion, and cluttered backgrounds

Note: Students should focus on developing a counting system that can accurately handle arbitrary object categories without retraining. The technical guidance serves as reference for project development.

9 Semantic Edge Detection

9.1 Project Background

Semantic edge detection identifies object boundaries by their semantic categories, unlike traditional methods that use low-level visual features. This is essential for advanced computer vision tasks like image segmentation and object recognition. The main challenge is to accurately detect both the precise location and the semantic meaning of edges in complex scenes.

9.2 Project Description

Students will design and implement a semantic edge detection system that takes an input image and produces edge maps where each edge pixel is assigned a specific semantic category label.

9.2.1 Requirements

- **Input:** RGB images of varying sizes
- **Output:** Semantic edge maps with categorical labeling
- **Core Functionality:** Must perform both edge detection and semantic categorization
- **Evaluation:** Qualitative evaluation on provided test dataset (see attachments)

9.3 Expected Outcomes

A successful project will deliver a functional semantic edge detection system demonstrating performance in:

- **Semantic Accuracy:** Meaningful categorization of edges by object boundaries
- **Edge Quality:** Clean, continuous edges with precise localization
- **Generalization:** Reasonable performance on unseen test data from provided dataset

9.4 Technical Guidance

9.4.1 Recommended Approaches

- **Multi-task Learning Framework:**
 1. Backbone network (ResNet, VGG) for feature extraction
 2. Parallel branches for edge detection and semantic segmentation
 3. Feature fusion for final semantic edge prediction
- **End-to-End Methods:** Architectures specifically designed for semantic edge detection (CASENet, etc.)

9.4.2 Critical Considerations

- Balance between edge detection sensitivity and semantic categorization accuracy
- Handle class imbalance in edge category distributions
- Maintain edge continuity while preserving semantic boundaries

Note: Students should focus on creating a semantically meaningful edge detection system. The project emphasizes qualitative evaluation on the provided test dataset, with model development conducted on standard public benchmarks. Technical guidance serves as reference for project development.

10 Transparent Object Segmentation

10.1 Project Background

Transparent object segmentation is a challenging computer vision task. Unlike ordinary objects, transparent materials lack consistent color or texture and visually merge with their backgrounds. This makes standard segmentation methods, which rely on these features, often fail. Robust transparent object segmentation is crucial for applications like robotic manipulation and augmented reality.

10.2 Project Description

Students will develop a system for segmenting transparent objects. The goal is to take an RGB image as input and generate accurate pixel-wise masks for all transparent object regions.

10.2.1 Requirements

- **Input:** RGB images containing transparent objects (glassware, bottles, windows, etc.)
- **Output:** Binary segmentation masks highlighting transparent object regions
- **Core Functionality:** Must handle various transparent materials and complex backgrounds
- **Evaluation:** Quantitative metrics on benchmark datasets and qualitative visual analysis

10.3 Expected Outcomes

A successful project will deliver an effective segmentation system demonstrating performance in:

- **Segmentation Accuracy:** High IoU (Intersection over Union) and F-score metrics
- **Boundary Precision:** Accurate edge localization for transparent objects
- **Robustness:** Consistent performance across different backgrounds and lighting conditions

10.4 Technical Guidance

10.4.1 Recommended Approaches

- **Encoder-Decoder Architecture:** Employ a U-Net-like structure to capture multi-scale contextual information in the encoder and progressively recover spatial details in the decoder to generate precise segmentation masks for transparent objects.
- **Attention Mechanism & Transformer:** Utilize self-attention and cross-attention mechanisms to model long-range dependencies and global context, which is critical for understanding the complex interactions between transparent objects and their background.
- **Multi-Task Learning:** Enhance the primary segmentation task by jointly learning auxiliary tasks such as boundary detection and depth estimation. This encourages the model to learn more robust and generalizable representations.

10.4.2 Dataset

Trans10k. (Xie et al. Segmenting Transparent Objects in the Wild.)

10.4.3 Critical Considerations

- Address the challenge of similar appearance between transparent objects and their backgrounds
- Handle various types of transparent materials with different optical properties
- Ensure the system can generalize to unseen transparent object categories

11 Self-Proposed Topic

11.1 Problem Background

Beyond the topics listed above, groups with a specific research interest or application idea not covered by the provided list may propose their own computer vision project.

11.2 Project Description

Design and implement a computer vision project of your own choosing. Your proposal should clearly define the problem, the input and output of your system, the methodology you intend to explore, and how you will evaluate the results. The scope and workload should be comparable to the ten predefined topics.

11.2.1 Requirements

- Clearly defined problem statement with well-specified input and output
- A concrete methodology grounded in computer vision techniques
- A sound evaluation plan with appropriate quantitative metrics and qualitative analysis
- Scope and workload comparable to the predefined topics